One fold, five allosteric architectures

Conservation and divergence in the allosteric architectures of five human protein kinases

Folgado, Beltran and Lehner — bioRxiv, 2026

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Central question: does a shared kinase fold imply a shared map of allosteric control?

Why compare allostery across kinases?

Kinases share a conserved ATP-binding cleft. This creates a difficult selectivity problem for inhibitors.

Alternative strategy
Perturb a secondary site that changes activity without directly occupying the catalytic site.

Two possibilities: all kinases retain the same functional secondary sites, or their common fold supports different regulatory architectures.

A structural pocket alone cannot tell us which possibility is true.

Introduction, pp. 2–3. Motivation, not a demonstration of drug selectivity.

Five kinases, with mutations tested in full-length proteins

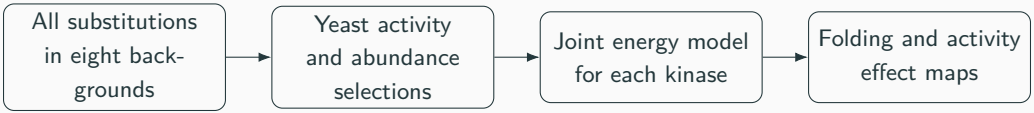
Kinase	Broad identity	Relevant context
SRC	Tyrosine kinase	Oncogenic signalling
FGR	SRC-family tyrosine kinase	Myeloid signalling
JNK2 / MAPK9	Serine/threonine kinase	Stress signalling
ZAK / MAP3K20	Serine/threonine kinase	Stress signalling; ZAKβ used
TSSK2	Serine/threonine kinase	Sperm maturation

Mutagenesis covers each kinase domain (269–357 residues); the assays express full-length constructs. SRC data come from a previous study.

Comparison unit: aligned kinase-domain positions, with regulatory context retained.

Results, p. 4; Extended Data Fig. 1l. Full-length domain architectures are shown in the Extended Data.

KINASE-MAPS: perturb, measure twice, then disentangle



All 19 substitutions at each domain site are designed in combinations with four substitutions at each of two background sites.

After quality control: **163,185 variants with activity measurements** and **231,716 with abundance measurements**.

Each single substitution is represented in a median of five double mutants.

Fig. 1C–D; Results, pp. 4–5; Extended Data Fig. 1E. Diagram redrawn.

Two growth readouts resolve a key confounder

Activity selection

Toxic kinase overexpression
More kinase activity → greater yeast toxicity → less growth.

Active-site mutations establish that toxicity depends on kinase activity.

Abundance selection

DHFR sandwich complementation
More folded protein → functional DHFR complementation → more growth under methotrexate.

This estimates folded abundance in the same cellular system.

Less toxicity could mean either less enzyme or less activity per folded molecule. Sequencing before/after selection measures variant enrichment.

Fig. 1D; Methods, pp. 16–20; Extended Data Fig. 1A–D.

A running example: the same activity readout has two causes

Illustrative variant	Folded abundance	Total activity	Interpretation
A	Reduced	Reduced	Could be a folding defect
B	Near wild type	Reduced	Activity-specific defect
C	Near wild type	Increased	Candidate activator

The joint model asks which combination of folding and activity effects explains *both* selections across multiple genetic backgrounds.

Why double mutants? Different backgrounds move a substitution through nonlinear response curves, helping constrain latent energy effects.

These are teaching examples, not measured variants.

Conceptual illustration based on Fig. 1H, Methods pp. 19–20, Extended Data Fig. 2A.

The inferred energies are useful, phenomenological parameters

Three model states: unfolded/inactive, folded/inactive, folded/active.

$$\Delta\Delta G_f = \text{change in folding energy}, \qquad \Delta\Delta G_a = \text{change in activity energy}.$$

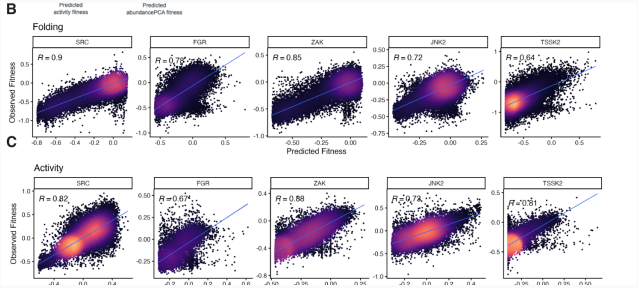
Sign	Folding parameter	Activity parameter
Positive	Destabilizing	Inactivating
Negative	Stabilizing	Activating

Single-mutation energy effects are assumed additive in double mutants; nonlinear transformations connect energies to fitness.

Interpretation: $\Delta\Delta G_a$ captures activity beyond abundance, including conformational shifts *and* catalytic changes. It is not a direct measurement of a unique active/inactive equilibrium.

Methods, pp. 19–21; Extended Data Fig. 2. Cross-kinase normalization uses reference mutations.

How reliable is the assay-to-energy inference?



Ten-fold cross-validation predicts held-out double-mutant fitness. Median Pearson r: 0.78 for abundance, 0.81 for activity. After attenuation correction: 0.85 and 0.90. Replicate correlations and stop/synonymous separation provide independent assay checks.

This validates prediction of assay fitness; it does not independently prove every latent energy or native signalling effect.

Extended Data Fig. 2B–C (cropped); Fig. 1E–H; Results, pp. 4–5.

Define an allosteric effect before mapping one

A mutation is called functional when its inferred effect exceeds 0.2 kcal/mol in magnitude and passes a one-sided z-test at BH FDR < 0.1.

An **allosteric mutation** is activating or inactivating and lies outside the defined active site.

The active site comprises 35 aligned positions: ATP contacts, magnesium-positioning residues, and substrate-positioning residues.

A **major allosteric site** is a residue enriched for activating or inactivating substitutions relative to the rest of that kinase.

Mutation calls, residue enrichment and pocket enrichment are different levels of evidence.

Methods, pp. 20–22; Results, pp. 6–7. Structural distances use ATP-bound AlphaFold3 models.

First result: fold stability is only partly conserved

Buried residues are enriched for destabilizing substitutions in all five proteins. Yet the lobe most sensitive to mutation varies:

SRC, FGR, ZAK	C-lobe enriched
JNK2	N-lobe enriched
TSSK2	Similar frequencies in both lobes

Per-position median folding effects correlate strongly for SRC–FGR ($r = 0.77$), less for SRC–JNK2 ($r = 0.40$) and SRC–TSSK2 ($r = 0.39$).

Even the buried αF helix shows protein-specific effects.

Implication: abundance correction cannot simply be transferred from a homolog.

Fig. 2; Extended Data Fig. 4C–G; Results, pp. 5–6. Correlations are position medians.

The catalytic machinery provides an internal biological check

Across five kinases: 3,996 inactivating and 1,087 activating substitutions are identified overall.

The active site is strongly enriched for inactivating mutations (pooled odds ratio 6.4). Magnesium-positioning and catalytic-loop contacts are especially sensitive.

The R-spine is mainly activity-sensitive: median $\Delta\Delta G_a = 0.53$ versus $\Delta\Delta G_f = 0.08$ kcal/mol.

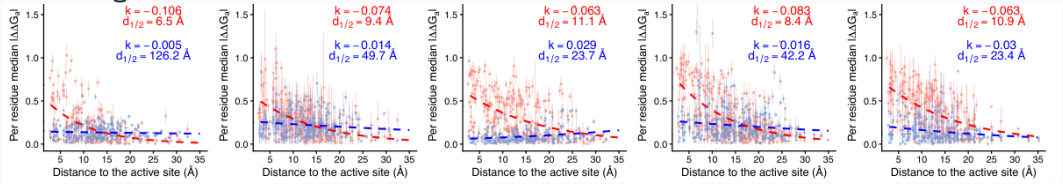
A conserved allosteric lever: the αC glutamate is the only position enriched for inhibitory allosteric mutations in all five proteins.

The adenosine-contact region is more divergent than the catalytic core.

Fig. 3A–H; Extended Data Fig. 5; Results, pp. 6–8. Position references use KinCore alignment.

Inhibition decays with distance; activation behaves differently

Left to right: SRC, FGR, ZAK, JNK2, TSSK2.



Red: mutations with $\Delta\Delta G_a > 0$; blue: $\Delta\Delta G_a < 0$. Each point is a residue median with standard-error bars; dashed curves are exponential fits.

$$|\Delta\Delta G_a| = a\, e^{kd}, \qquad d_{1/2} = \ln(2)/|k|.$$

Inhibitory half-distances span 6.5–11.1 Å. Activating effects show no comparable enrichment near the active site.

Distance explains only about 29% of inhibitory-effect variance on average.

Fig. 4B, cropped below repeated preprint header; Results, pp. 8–9. Distance is to ATP or catalytic Asp.

Allostery also depends on direction

Reference structure, then SRC, FGR, ZAK, JNK2, TSSK2.

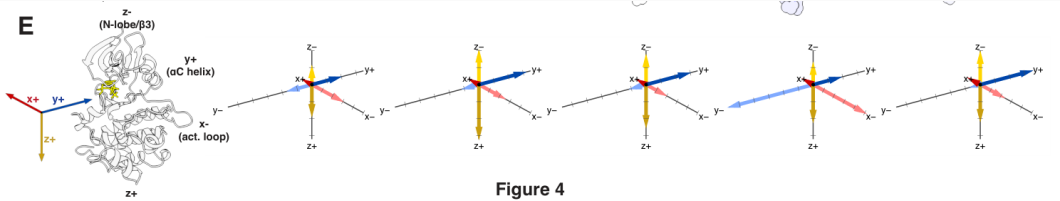


Figure 4

Arrow length encodes inhibitory half-distance; ticks are 10 Å. Longer arrows indicate slower decay.

Coupling is strongest toward αC (median $d_{1/2} = 26.1$ Å) and weaker toward αD (4.8 Å).

JNK2 couples especially strongly toward the activation loop: 46.4 Å versus 14.9 Å in ZAK.

Interpretation: a shared distance-decay principle coexists with protein-specific spatial wiring.

Fig. 4E (cropped); Methods pp. 21–22. Directional fits use restricted coordinate slabs, not molecular paths.

Activation is especially idiosyncratic

Outside the active site, the study reports 2,617 inhibitory and 1,040 activating mutations across the five kinases.

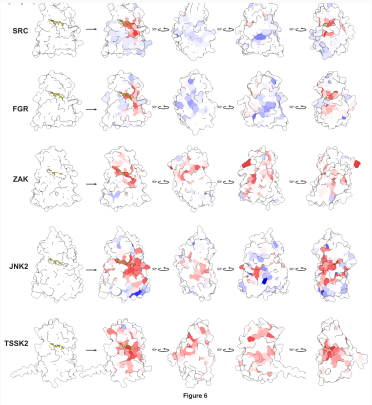
Of 47 aligned positions enriched for activating mutations, **43 are enriched in only one kinase**. Seven sites activate one kinase but inhibit another.

Of 946 distinct activating substitutions (aligned site + mutant amino acid), 860 activate just one kinase; none activate all five.

Conserved fold, divergent control
The common architecture constrains catalysis more strongly than it constrains the locations where activity can be increased.

Fig. 5; Extended Data Fig. 6G and 7A; Results, pp. 9–11. Counts pool kinase-specific measurements.

The protein surface is where divergence becomes useful



Rows: SRC, FGR, ZAK, JNK2, TSSK2. Four rotated surface views follow an uncoloured reference.

Blue = negative median $\Delta\Delta G_a$; red = positive; white near zero.

SRC/FGR have similar surfaces.

ZAK/TSSK2 are inhibition-biased. JNK2 has both signs.

Caution: a near-neutral median can hide a mixture of strong activating and inhibitory substitutions.

Fig. 6, cropped (top header omitted); Results, p. 11. Original colour scale: -1 to +1 kcal/mol.

From residues to candidate drug pockets

FTMap docks small chemical probes onto ATP-free lobes of each structural model. Probe-contacting residue sets are merged across aligned kinases into **37 global pockets**.

For each pocket, ask two questions:

1. Is it enriched for activating or inhibitory mutations relative to the rest of that kinase?
2. Is its activity-effect distribution different from the homologous pocket in other kinases?

The second test subtracts each kinase's domain-wide median first, then uses pairwise Wilcoxon tests with BH correction.

Why this matters: enrichment in one kinase and non-significance in another alone do not establish specificity.

Methods, p. 22; Fig. 7A; Extended Data Fig. 8C–D. Druggability score = number of probe clusters.

Shared inhibitory pockets coexist with selective candidates

Conserved coupling	DFG (13), MT3 (15), and pockets 19/20 are inhibitory across all five; they cluster near the interlobe cleft.
JNK2-specific candidates	DRS (23) and DEF (30): substrate-recruitment surfaces passing the stricter specificity criterion.
SRC-specific candidate	Pocket 1 near αC/β4; differs even from close homolog FGR.

SRC and FGR retain highly correlated landscapes, but candidate pocket differences remain.

These are experimentally supported mutational targets, not demonstrated selective small-molecule inhibitors.

Fig. 7A,D; Extended Data Fig. 8D; Results, pp. 12–13. Specificity is based on baseline-corrected pairwise tests.

A pocket can change regulatory sign across homologs

Pocket	Activating enrichment	Inhibitory enrichment
21	SRC	ZAK, JNK2, TSSK2
26 (αE/αF face)	SRC, FGR, JNK2	ZAK
33 (DEF)	SRC, ZAK	JNK2

Seven pockets show opposite signs across kinases. Some pockets also contain both activating and inhibitory perturbations *within* one protein.

In TSSK2 pocket 2, one activating position sits among several inhibitory positions.

Consequence: which residue and chemistry a ligand perturbs may matter as much as where it binds.

Fig. 7A,D,E; Extended Data Fig. 8C–D; Results, p. 14. Table summarizes mutation enrichments, not ligand responses.

What the study establishes, and what it leaves open

Established in this assay

- Abundance-adjusted activity maps for five full-length kinase constructs.
- Conserved, anisotropic inhibitory distance decay.
- Divergent activation sites and functional pocket repertoires.

Still requires validation

- Native mammalian signalling, substrates and phosphorylation contexts.
- Ligands that reproduce mutational effects and achieve selectivity.
- Generality across the >500-member human kinome.

Additional limits: additive latent energies; proxy growth phenotypes; reference-based energy normalization; predicted structures; significance thresholds and incomplete measurements.

Discussion, pp. 15–16; Methods, pp. 19–22. Right column and synthesis are journal-club interpretation.

A concrete next experiment and a discussion question

Test the prediction, not just the map.

Choose the SRC-specific pocket 1. Screen or design binders, verify binding, then measure purified SRC and FGR activity at matched ligand occupancy. Follow with mammalian target engagement and signalling assays.

A compound that binds both proteins but inhibits only SRC would support functional-coupling selectivity; selective binding would support a different mechanism.

Discussion
How much of the apparent divergence is encoded by the kinase domain, and how much comes from its full-length regulatory context or the yeast assay?

Take-home: the same fold supports shared catalytic constraints and distinct routes to regulation.

Proposed experiment and interpretation. Sources: Fig. 7, Discussion; DOI 10.64898/2026.08.04.742685.